Check whether the zero marking of the main scale coincides with that of the zero of the vernier scale. If they coincide then there is no zero error. If they do not coincide with each other, the instrument is said to possess zero error. Zero error may be positive or negative. If the zero of a vernier is shifted to the right of main scale, it is called positive error. On the other hand, if the zero of the vernier is shifted to the left of the zero of main scale, then the error is negative.

Positive zero error

Figure 1.3 shows the positive zero error. From the figure you can see that zero of the vernier scale is shifted to the right of the zero of the main scale. In this case the reading will be more than the actual reading. Hence, this error should be corrected. In order to correct this error, find out which vernier division is coinciding with any of the main scale divisions. Here, fifth vernier division is coinciding with a main scale division. So, positive zero error = +5 × LC = +5 × 0.01 = 0.05 cm and the zero correction is negative. Hence, zero correction is –0.05 cm.

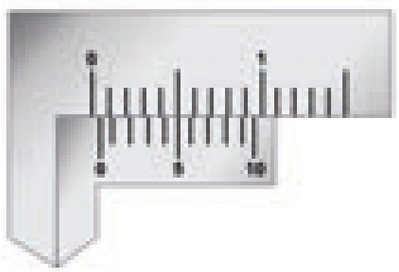


Figure 1.3 Positive zero error

Problem 1

Calculate the correct reading, if the main scale reading is 8 cm, vernier coincidence is 4 and positive zero error is 0.05 cm.

Solution:

Correct reading = 8 cm + (4 × 0.01cm) – 0.05 cm
= 8 + 0.04 – 0.05 = 8 – 0.01 = 7.99 cm

Negative zero error

Now look at the Figure 1.4. You can see that the zero of the vernier scale is shifted to the left of the zero of the main scale. So, the obtained reading will be less than the actual reading. To correct this error we should first find which vernier division is coinciding with any of the main scale divisions, as we found in the previous case. In this case, you can see that sixth line is coinciding. To find the negative error, we can count backward (from 10). Here, the fourth line is coinciding. Therefore, negative zero error = –4 × LC = –4 × 0.01 = –0.04 cm. Then zero correction is positive. Hence, zero correction is +0.04 cm.

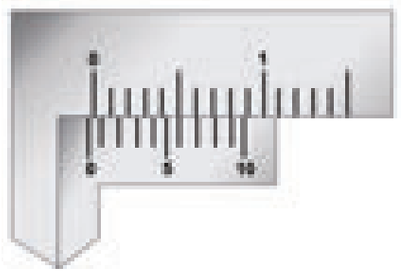


Figure 1.4 Negative zero error

Problem 2

The main scale reading is 8 cm and vernier coincidence is 4 and negative zero error is 0.02 cm. Then calculate the correct reading:

Solution:

Correct reading = 8 cm + (4 × 0.01 cm) + (0.02 cm)
= 8+0.04+0.02 = 8.06 cm.

We can use Vernier caliper to find different dimensions of any familiar object. If the length, width and height of the object can be measured, volume can be calculated. For example, if we could measure the inner diameter of a beaker (using appropriate jaws) as well as its depth (using the depth probe) we can calculate its inner volume.

Activity 1

Using Vernier caliper find the outer diameter of your pen cap.



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